Experiment - 7

Sliding window protocol

Aim:

To write a program to implement flow control at data link layer using sliding window protocol simulate the flow of frames from one node to another

Sender code:

```
import time
import os
print ("=",55)
print ("sliding window protocol - sender")
window_size = int (input ("Enter window size"))
if (window_size <=0;
    print ("Invalid window size!")
    exit ()
message = input ("Enter text message:")
if not message:
    print ("message cannot be empty!")
    exit()
frames = []
frame = []
for i, character in enumerate (message);
    frames, append ("frames_no"; i, "data", character
})
print ("Frames created :")
Print ("['frame no : {frame ["frame_no"]} DATA : {frame['data']}
total frames = len (frames)
base = 0
```

```
while base = label.frames
    end = min (base + window size, label - frames)
    print ("sending frames {base} to find")
    with open ("sender - buffer", w) as file; file i in range
                                        (base end ()
else :
    print ("Invalid response recieved time")
    time.sleep(1)
    print ("All frames transmitted successfully")
print ("original message {message}")
```

output :

```
SLIDING WINDOW PROTOCOL SENDER
Enter window size : 3
Enter test message : HELLO
Sender buffers
Frame NO : 0        DATA : H
Frame no : 1        DATA : E
Reciever response : ACK 3
all frames transmitted successfully!
Original message : HELLO
```

Reciever code :

```
Import OS
SENDER_FILE = "sender_buffer.txt"
RECIEVER_FILE = "reciever_buffer.txt"
expected frame = 0
while true :
    if not os.path.exsit (sender file):
```

```
else:
    with open (RECEIVER_file, 'w') as file;
        file.write (f"Ack : {expected file}")
    print ("\n ALL frames recieved correctly ")
    print (f"ACK : {expected_frame}")
```

Output: Frames Recieved

frame NO : 0 Data : H

frame NO : 1 Data : E

All frames recieved correctly

Ack : 3

Result :

Thus the sliding window protocol was implemented successfully using Python and flow of frames btw sender and reciever was simulated using ACK/NACK.

Experiment - 8

NMAP TO DISCOVER LIVE HOSTS USING NMAP SCANS

Aim:

To discover live host on a network using Nmap host - discovery scans - ARP scan, ICMP scan, and TCP/UDP Ping scan - and to understand which protocol operates at which layer of TCP/IP model, when each one applies

Nmap:

Nmap is a free open - source network scanning tool created by Gorden Lyon and first released in 1997

Procedure:

1) log onto Tryhackme and open the nmap live host discovery room

2) Work through tasks 1-3 and answer the simulator based question

3) Deploy the lab network in Task is and start the attackbox note target subnet

4) Run in ARP scan. PR <subnet>/24 and record how many host respond

5) Run an ICMP echo scan. Sudo nmap - PE result with ARP Scan.

6) Run ICMP timestamp and ICMP address.

7) Run TCN SYN (Ps) TCP ACK, UDP scan

8) Complete the knowledge check que and proceed to nmap base port scan room.

Experiment - 9

Implementation of subnetting in CISCO PACKET TRACER Simulator

Aim:

To implement subnetting in Cisco Packet Trace. by creating multiple subnet assign static IP address connecting the subnets using routes, testing Packet transmission

Theory:

Subnetting is the process of dividing one network into smaller network called subnet. It helps in efficient use of IP address and allow multiple network to be created from a single network address.

Requirements:

Cisco packet Trace
Router
Switches
PCs
Copper Straight - Through Cables

Procedure:

1) open cisco packet Trace and create a new network topology

2) Add the required router - switches and PCs to the workspace.

3) Connect the PCs to switch to the routers using suitable cables.

4) Divide
Subnet

5) Assign
the defa

6) Contig

7) Cont
gateway

8) Chec

9) Sen

10) Ch
Route
Device
Router
PC1
PC2
PC3

4) Divide the network 192.168.1.0124 units smaller subnet using the 127 subnet mask.

5) Assign static IP address to all PCs and configure the default gateway

6) configure the router interface through the CL

7) Configure IP address, subnet mask and default gateway on each PC

8) Check the connection blw PCs using Ping command

9) Send a simple PDU blw PCs in diff subnet

10) observe Packet transmission with and without Route

Device	IP address	subnet mask	Gateway
Router	192.168.1.1	255.255.255.224	-
PC1	192.168.1.11	255.255.255.224	192.168.1.1
PC2	192.168.1.12	255.255.255.224	192.168.1.1
PC3	19.2.168.1.13	255.255.255.224	192.168.1.1

Result :

Thus, subnetting was successfully implemented in Cisco packet Tracer.

Experiment - 10

Internetworking with routers in Cisco Packet tracer simulator

Aim:

To design and configure a simple Internetwork using a router and establish communication blw two diff network.

Requirement:

Cisco packet tracer

1. Routis
2. PCs
3. copper straight through cables.

Theory:

A router is a networking device used to connect diff network and forward packet blw them in two pcs are connected to two diff interface of router static IP address are assigned to PCs and route interface.

IP configuration:

Device	IPaddress	subnetwork	gateway
Routis Fa010	192.168.10.1	255.255.255.0	-
Router Fa 011	192.168.20.1	255.255.255.0	-
PC0	192.168.10.2	255.255.255.0	192.162.1.0.1
PC1	192.168.20.2	255.255.255.0	192.162.2.0.1

Procedure:

1) Open Cisco packet Tracker
2) Place one router and two PCs in workspace
3) Connect PC0 to Router Fast Ethernet 0/0 using copper straight through cable
4) Connect PC1 to Router, Fast Ethernet 0/1, using copper straight through cable
5) Open router CLI and enter
6) Configure PC0

   IP: 192.168.10.2
   Subnet mask: 255.255.255.0
   Default gateway: 192.168.10.1

7) Open command prompt PC0 and type ping 192.168.20.2
8) Observe the reply packet
9) Send a simple PC0 from PC0 to PC1 and observe the packet transmission.

Result:

Thus the network was successfully designed and configured using a router in Cisco packet Tracker.

Experiment - 11

Routing at Network Layer

Aim

To simulate static Routing and RIP using CISCO packet tracer

A) Static Routing:

Static Routing is a method where routes are manually added to the router routing table It is useful for controlling path taken by packet d cv diff network.

Procedure:

1) Open Cisco Packet Tracer
2) Place required router, Switch and PC in workspace
3) Connect the device using suitable cables.
4) assign IP address to all PCs and router interfaces
5) Open the router CLI
6) Enter the full command to configure static router
7) Use full command to check routing table show IP route
8) Use ping command to check connectivity btw Pcs

B) RIP Routing:

RIP is a dynamite routing protocol. It allows routers to automatically exchange information abt the network connected to them. RTP selected router based mainly on no. of hops.

Procedure:

1) Create network with three routes and PCs shown in packet trace topology
2) Assign IP address to PCs and routes interface
3) Configure the routes interface using
4) Enable RIP on routes
5) Repeat RIP configuration for other route with their directly connected ~~be~~ network.
6) Allow some time for routes to exchange routes with their directly connected network.
7) Send simple PDV from source PC to destination PC and observe packet path.

Observation:

In static routine route are manually configured by addmission in DIP routes automatically exchange routing into.

Result

succe

d PCs

interface

te with

routes

ination

ntioned

f exchange

Result:

Thus, static routine and RID were successfully simulated using Cisco packet Tracker.

Experiment-12

End to end communication at transport layer.

Aim:

To Implement an echo-client server using TCP/UDP socket and Implement a chat client servers using socket programming.

The client sends a message to server and the server sends the same message back to client.

Procedure:

1) create server program and clients program
2) Import the required socket library
3) In server creates a socket and mind it to an IP address and port.
4) Make the server listen to the client request
5) start client and connect it to server
6) enter msg from client
7) Send msg to server
8) The seuce recieves the msg and send same msg back.

b) client

1) create
program

2) start
and po

3) make

4) start

5) Enter

6) The

Result:

The

Implem

b) client server using socket progress:

1) create server and client program using socket programming

2) start server and bind it to an IP address and port

3) make server wait for client connection

4) start client and connect it to server

5) Enter msg to client

6) The server send a reply back to client.

Experiment - 13

Implement your own ping program

Aim:

To Implement a simple ping program to check the connectivity blw two network devices.

Theory:

Ping a network utility used to check wether a destination device is reachable through a network and waits for a reply - It reply is received connection is successful.

Requirements:

Computer

Python / suitable programming environment

Network connection.

Procedure:

1) Create a new Program for implementing ping operation.
2) Import the required socket / network library
3) Get IP address or hostname of the destination
4) Create a socket for communication
5) Send req packet to the destination
6) wait for response is recieved ~~pls~~ display that destination is reachable

Algorithm:

1) Start
2) Enter
3) Create
4) Send
5) wait
6) If
8) Stop

Result

Imple

Algorithm:

1) Start
2) Enter destination IP address
3) Create a socket
4) Send a ping request
5) wait for response
6) If response is recieved
8) stop.

Experiment - 14

Raw sockets - packet snifting

Aim:

To write a program using RAW socket to Implement packet snifting

Theory:

A packet snifter is a program used to capture and examine network packets directly at lower level, so packet information such as source address destination address.

Requirements:

Complete / Laptop
Python
Raw Socket support
network Interface
administrator / Root privileges.

Algorithm:

1) start the program
2) Create RAW socket
3) Find the socket to required network
4) Wait for incoming network packet
5) Capture the packet using RAW socket
6) Extract packet Information
7) display packet detail.
8) Stop pgm when required

Procedure:

1) Open
2) Create
sniffing
3) Import
4) Create
5) Displ
6) Open
the ca

Result:

Thus
was Imp

Procedure:

1) Open Python IDE or terminal
2) Create a python pgm for packet sniffing
3) Import socket module
4) Create RAW socket
5) Display Packet Information
6) Generate some network traffic observe the captured packets.

Result:

Thus a Raw socket based Packet filling Pgm was Implemented successfully.

Experiment - 15

Analysis of weblogs using webalizer tool.

Aim:

To analyse the diff types of web logs using webalizer tool.

Theory:

webalizer is a web log analysis tool used to process websecure log files and generate reports

Requirements:

- Computer / Laptop
- webalizer window version
- web log file

Procedure:

1) Run the webalizer window version
2) Select or provide the requirements web log file
3) download web log file from web if required
4) load the web-log file into webalizer
5) Press Run webalizer
6) webalizer process the log file
7) Observe the generates analysis / response
8) Study the diff information available in the report

Input:

A

output:

Web

webalizer.

Result:

Thus

analyzed

Input:
A web servers log files.

output:
web log analysis report generats using webalizer.

Result:

Thus the web log file was sucessfully analyzed using webalizer tool.